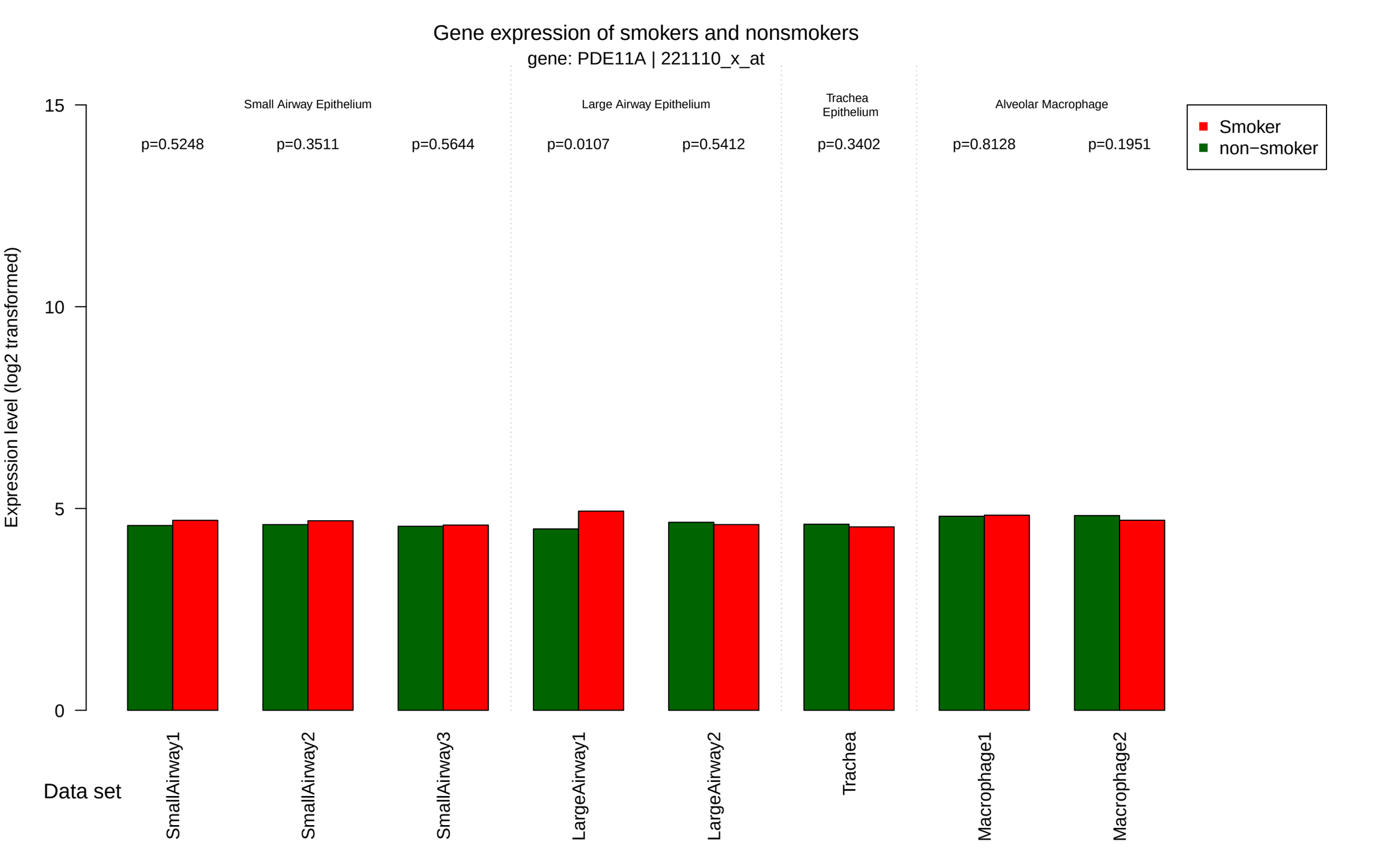
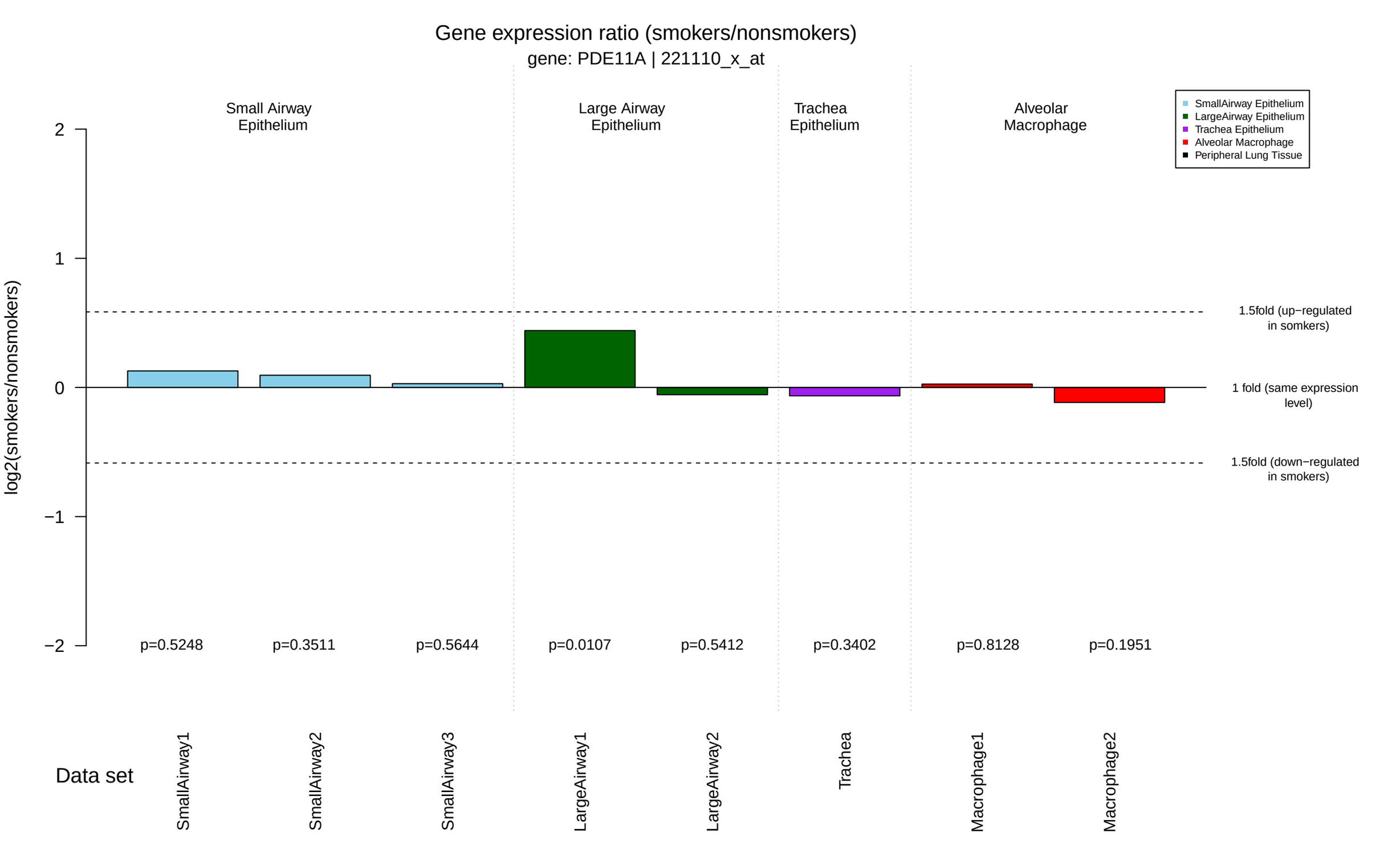
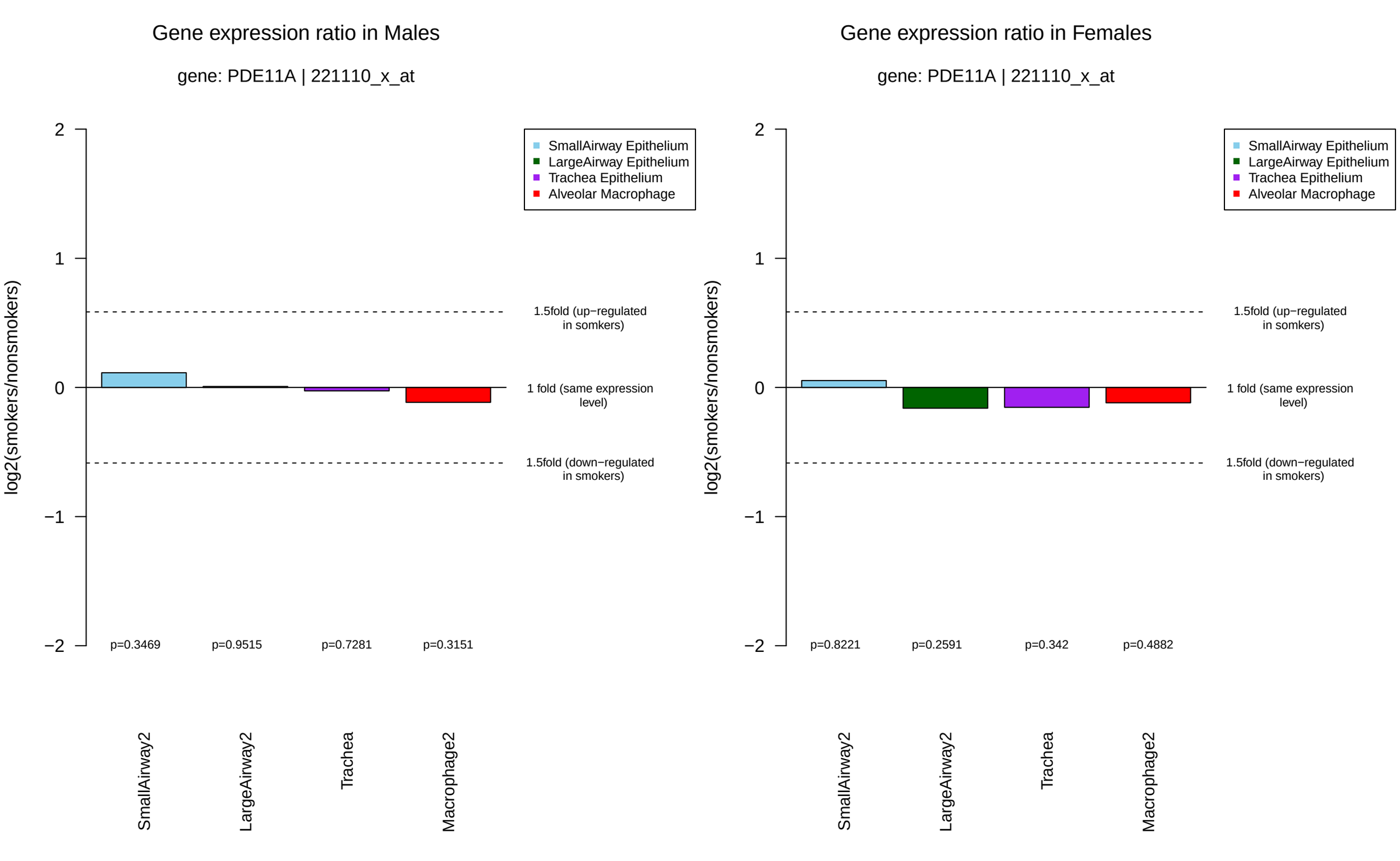
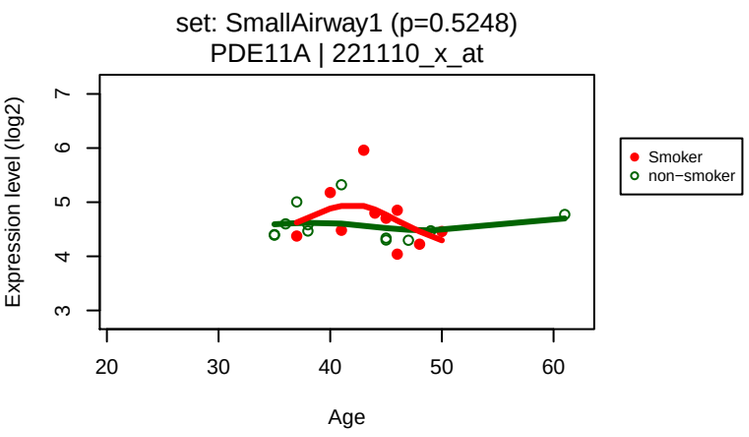
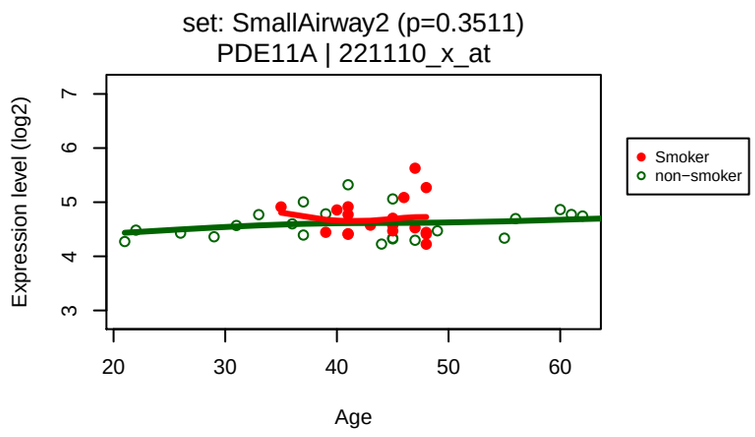
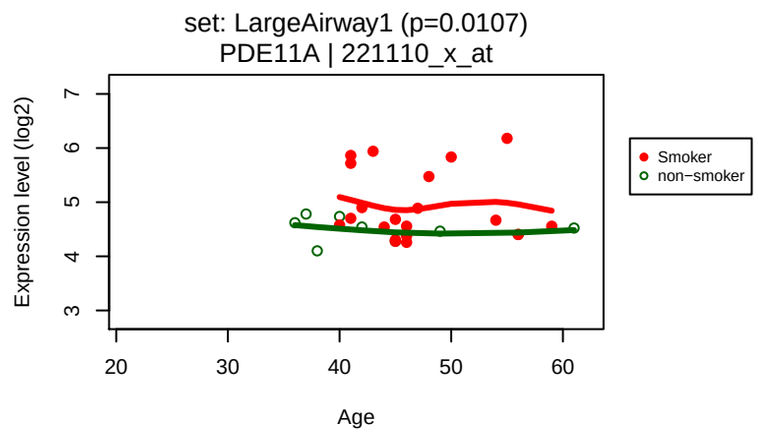

Gene expression of smokers and nonsmokers
gene: PDE11A | 221110_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5248
p=0.3511
p=0.5644
p=0.0107
p=0.5412
p=0.3402
p=0.8128
p=0.1951
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

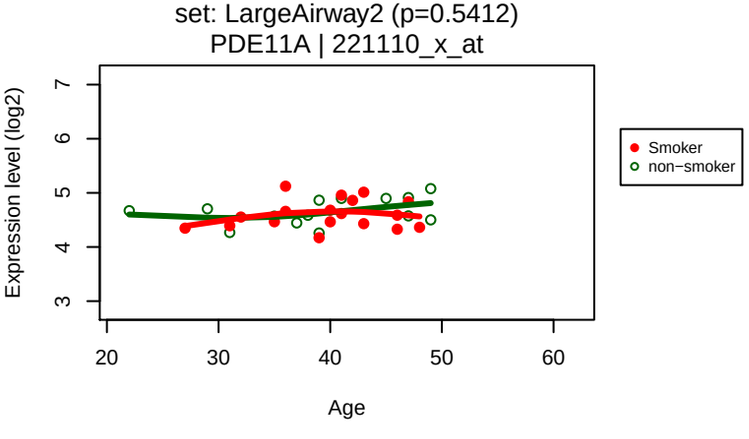

Gene expression ratio (smokers/nonsmokers)
gene: PDE11A | 221110_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5248
p=0.3511
p=0.5644
p=0.0107
p=0.5412
p=0.3402
p=0.8128
p=0.1951
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

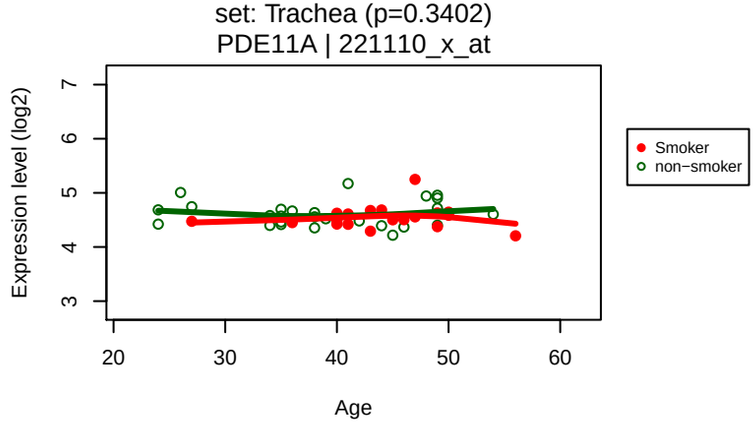

Gene expression ratio in Males
gene: PDE11A | 221110_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3469
p=0.9515
p=0.7281
p=0.3151
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE11A | 221110_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8221
p=0.2591
p=0.342
p=0.4882
SmallAirway2
LargeAirway2
Trachea
Macrophage2

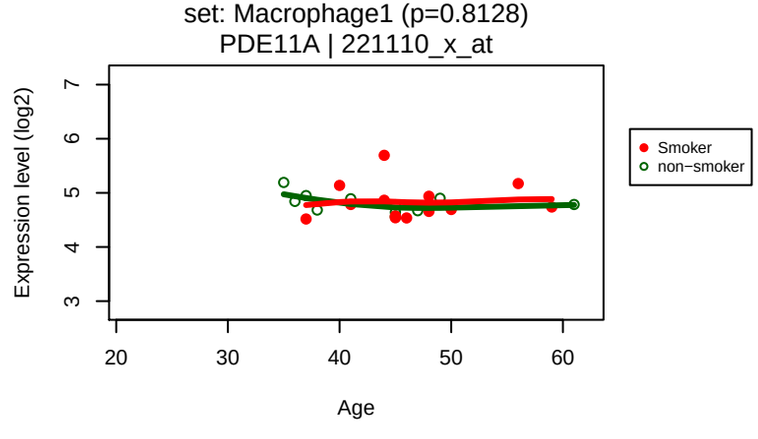

set: SmallAirway1 (p=0.5248)
PDE11A | 221110_x_at
Expression level (log2)
Age
Smoker
non−smoker

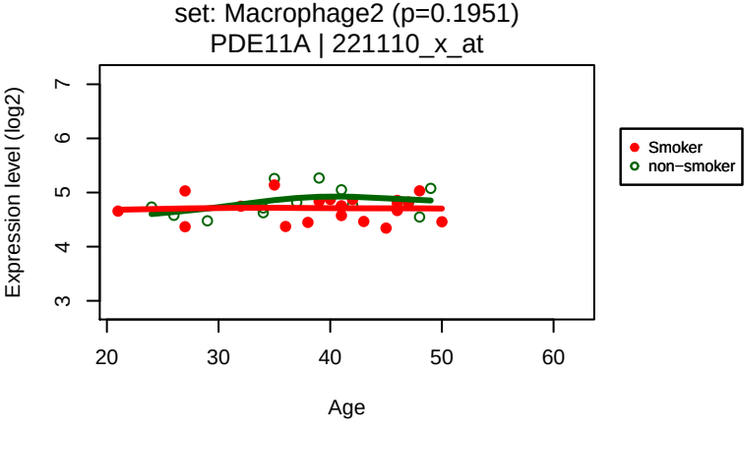

set: SmallAirway2 (p=0.3511)
PDE11A | 221110_x_at
Expression level (log2)
Age
Smoker
non−smoker

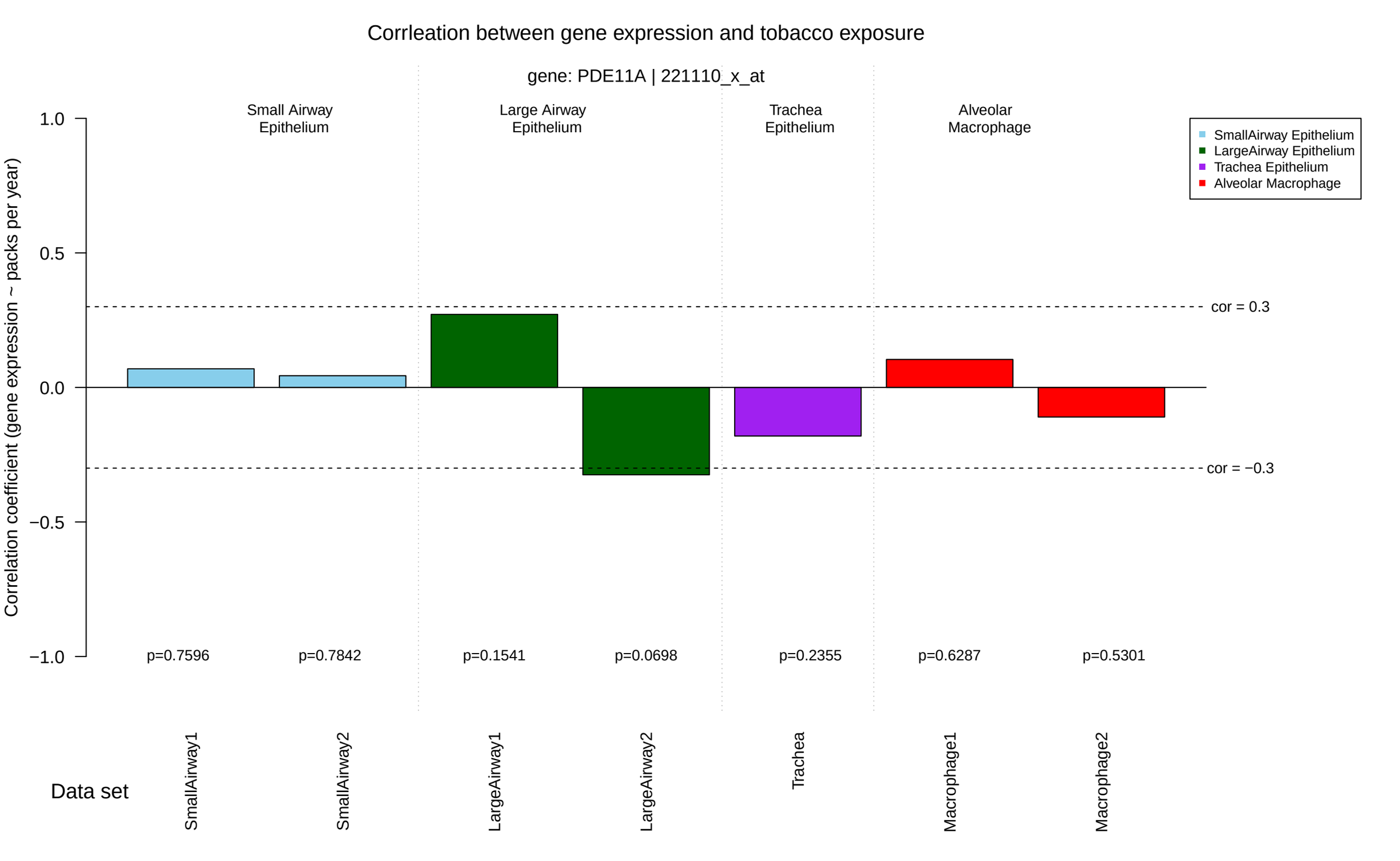

set: LargeAirway1 (p=0.0107)
PDE11A | 221110_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5412)
PDE11A | 221110_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3402)
PDE11A | 221110_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8128)
PDE11A | 221110_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1951)
PDE11A | 221110_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE11A | 221110_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7596
p=0.7842
p=0.1541
p=0.0698
p=0.2355
p=0.6287
p=0.5301
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2